AIR INTAKE MODE DOES NOT CHANGE [MANUAL AIR CONDITIONER]

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RECIRCURATE/FRESH Lever Type Control Unit

DESCRIPTION	• Air intake mode does not change when switching REC/FRESH mode.
POSSIBLE CAUSE	• Air intake wire, clamp malfunction • Climate control unit air intake control lever malfunction • A/C unit air intake door malfunction

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY AIR INTAKE WIRE FROM A/C UNIT IS POSITIONED SECURELY AND CORRECTLY** • Is the air intake wire securely installed in the correct position in relation to the A/C unit air intake links?	Yes	Go to the next step.
		No	Adjust the air intake wire or install securely in the correct position, then go to Step 5.
2	**INSPECT CLIMATE CONTROL UNIT** • Remove the air intake wire from A/C unit. • Inspect climate control unit air intake control lever operation. • Does the air intake lever move smoothly?	Yes	Go to the next step.
		No	Repair or replace the climate control unit, then go to Step 5.
3	**INSPECT AIR INTAKE DUCT** • Is there any foreign material or obstruction in the air intake door?	Yes	Remove obstruction, then go to Step 5.
		No	Go to the next step.
4	**INSPECT AIR INTAKE DOOR** • Is the air intake door securely and properly installed?	Yes	Inspect the air intake door for cracks or damage, then go to the next step.
		No	Install the air intake door securely in the proper position, then go to the next step.
5	**CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR** • Does the air intake mode change smoothly?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if the malfunction recurs.

RECIRCURATE/FRESH Switch Type Control Unit

DESCRIPTION	• Air intake mode does not change when switching REC/FRESH mode
POSSIBLE CAUSE	• Air intake actuator malfunction • Open or short circuit in wiring harness between climate control unit and air intake actuator • Climate control unit malfunction • Air intake link malfunction • Air intake door malfunction • Restriction in blower unit

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, inspect to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Note

- The climate control unit controls the air intake mode when the airflow mode is in DEFROSTER so that it is fixed at FRESH mode. At this time, the airflow mode cannot be changed from FRESH to RECIRCULATE even if the REC switch is pressed, but this does not indicate a malfunction.

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT TO SEE WHETHER MALFUNCTION IS IN AIR INTAKE ACTUATOR OR ELSEWHERE** • Disconnect the air intake actuator connector. • Switch the ignition ON (engine off or on). • Measure the voltages at the following climate control unit terminals. **L.H.D.** — Terminal F (FRESH motor drive signal) — Terminal A (RECIRCULATE motor drive signal) **R.H.D.** — Terminal A (FRESH motor drive signal) — Terminal F (RECIRCULATE motor drive signal) • Are voltages normal?	Yes	Go to the next step.
		No	Go to Step 3.
2	**INSPECT AIR INTAKE ACTUATOR** • Inspect air intake actuator. (See AIR INTAKE ACTUATOR INSPECTION [MANUAL AIR CONDITIONER].) • Is it normal?	Yes	Go to the Step 6.
		No	Replace the air intake actuator, then go to Step 8. (See AIR INTAKE ACTUATOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)
3	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO POWER SUPPLY BETWEEN CLIMATE CONTROL UNIT AND AIR INTAKE ACTUATOR) OR ELSEWHERE** • Air intake actuator connector disconnected. • Disconnect the climate control unit connector. • Measure the voltages at the following air intake actuator terminals. — Terminal A — Terminal B • Are voltages **approx. 0 V**?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air intake actuator terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
4	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO GROUND BETWEEN CLIMATE CONTROL UNIT AND AIR INTAKE ACTUATOR) OR ELSEWHERE** • Climate control unit and air intake actuator connectors disconnected. • Switch the ignition off. • Inspect for continuity at the following terminals between the air intake actuator and ground. — Terminal F (L.H.D.) / A (R.H.D.) (FRESH motor drive signal) — Terminal A (L.H.D.) / F (R.H.D.) (RECIRCULATE motor drive signal) • Is there continuity?	Yes	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air intake actuator terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.

Step	Inspection		Action
5	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY OR OPEN BETWEEN CLIMATE CONTROL UNIT AND AIR INTAKE ACTUATOR) OR ELSEWHERE** • Climate control unit and air intake actuator connectors disconnected. • Inspect for continuity at the following terminals between the air intake actuator and climate control unit. **L.H.D.** — Terminal F—A (FRESH motor drive signal) — Terminal A—B (RECIRCULATE motor drive signal) **R.H.D.** — Terminal A—A (FRESH motor drive signal) — Terminal F—B (RECIRCULATE motor drive signal) • Is there continuity?	Yes	Replace the climate control unit, then go to Step 8. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air intake actuator terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to Step 8.
6	**INSPECT AIR INTAKE LINK** • Inspect the air intake link. (See BLOWER UNIT REMOVAL/ INSTALLATION.) — Is there grease on link? — Are the links securely and properly installed? — Are the links free of obstructions and hindrances? • Are the above items normal?	Yes	Go to the next step.
		No	Apply grease to the links. If any the links are damaged, replace the air intake actuator, then go to Step 8. (See BLOWER UNIT DISASSEMBLY/ASSEMBLY.)
7	**INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR AIR INTAKE DOOR** • Inspect the blower unit air intake door. (See BLOWER UNIT REMOVAL/ INSTALLATION.) — Is the door free of obstructions, cracks, and damage? — Are the doors securely and properly installed? • Are the above items normal?	Yes	Go to the next step.
		No	Remove obstruction, or install the doors in the proper position. If any doors are cracked or damaged, replace them, then go to the next step.
8	**CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR** • Does the air intake mode change smoothly?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if the malfunction recurs.